



Efficient Threshold Voltage Adjustment Technique by Dielectric Capping Effect on MoS₂ Field-Effect Transistor

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Abstract—In this letter, we demonstrate an efficient threshold voltage (V_{th}) adjustment technique by depositing single or double dielectric layers on MoS₂ field-effect transistors (FETs). We used Al₂O₃ and SiO₂ as the capping layers on the MoS₂ FET and observed different average V_{th} shifts of -2.39 and $+7.13$ V, respectively. In order to further the controllability of the dielectric capping effect, the deposition of double dielectric layers, specifically Al₂O₃ on SiO₂ and vice versa, was used for the first time. Consequently, the deposition of SiO₂ on Al₂O₃ and Al₂O₃ on SiO₂ shows an average V_{th} shifts of $+4.92$ and $+4.02$ V, respectively. The defect charges in the dielectric layer can induce band bending in the MoS₂ interface and adjust V_{th} of the MoS₂ FET. This result suggests a promising V_{th} adjustment technique for transition-metal dichalcogenides-based FETs.

Index Terms—MoS₂, field effect transistor, threshold voltage.

I. INTRODUCTION

RESEARCH on two-dimensional (2-D) layered structure transition-metal dichalcogenides (TMDC) has been attempted to overcome the current limits of silicon-based complementary metal-oxide-semiconductor (CMOS) technology [1]–[5]. Recently, TMDCs such as molybdenum disulfide (MoS₂) and tungsten diselenide (WSe₂) have been in the spotlight because they were found to have a band-gap and show monolayer properties [6], [7]. There are several advantages of TMDC over conventional semiconductors. First, TMDC has no surface dangling bonds due to its van der Waals epitaxial structure and hence, free from the carrier mobility degradation caused by surface oxidation [1]. Second, the short channel effect does not exist owing to its layered structure [8]. Consequently, TMDC-based devices can

achieve near ideal subthreshold swing and remarkable carrier transport properties [9]. Moreover, the energy band-gap of some TMDCs is tunable (1.2 eV $\sim$ 1.8 eV) by changing the number of layers [10]. Therefore, TMDCs are considered as the promising materials for optoelectronic devices [11].

While threshold voltage (V_{th}) adjustment is very important for device power consumption and performance [12], in a TMDC-based device, it is very hard to control V_{th} due to the atomically thin nature of TMDC [13]. In conventional CMOS technology, V_{th} can be controlled by modulating the active doping level of the channel region. However, general ion implantation is inappropriate for TMDC because the technique can induce fatal damage to its crystal structure [14]. Thus, several research groups have attempted to develop a proper V_{th} adjustment method for TMDC-based field-effect transistors (FET) [15]–[17]. Recent research has demonstrated that the deposition of a dielectric layer on top of the TMDC channel can adjust V_{th} in TMDC-based FETs. Na *et al.* adopted Al₂O₃ as a dielectric layer on a MoS₂ FET and showed that a change in V_{th} was induced by the positive fixed interface charges in the dielectric layer [18]. Liu *et al.* [19] showed the change in V_{th} caused by using Al₂O₃ on a phosphorene transistor. While these researches achieved V_{th} shifts in TMDC-based FETs by using a dielectric layer, they used only Al₂O₃ as the dielectric layer and could achieve only a negative V_{th} shift. Therefore, the development of a new technique based on the dielectric layer is necessary to obtain efficient V_{th} adjustment for high performance TMDC-based FETs.

In this letter, we investigated the dielectric capping effect of single or double dielectric layers on V_{th} adjustment of a MoS₂ FET. Single and double dielectric layers were deposited on the MoS₂ FET, and subsequently the shift of V_{th} of the MoS₂ FET was extracted. Furthermore, the distribution of the shift of V_{th} was investigated to exclude variation of procedure. These results could be explained by band diagrams, especially the band bending of MoS₂ induced by the defect charges in the dielectric layer.

II. EXPERIMENTAL PROCEDURE

MoS₂ flakes were mechanically exfoliated from bulk MoS₂ (SPI Supplies Inc.) using a simple tape peeling method, and then transferred to a 300nm thick thermally grown SiO₂ layer on a heavily doped p-type silicon substrate ($0.005\Omega \cdot \text{cm}$). The measured thickness of the MoS₂ flakes is $\sim 30\text{nm}$

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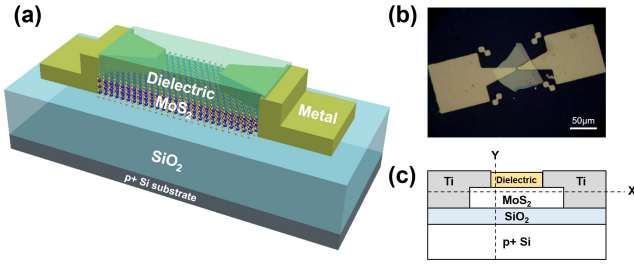


Fig. 1. (a) 3D-Schematic diagram of a back-gated MoS₂ field effect transistor (FET). Source/drain contacts are Ti(30nm)/Au(40nm). (b) Optical microscope image of the back-gated MoS₂ FET. (c) Device structure of the back-gated MoS₂ FET.

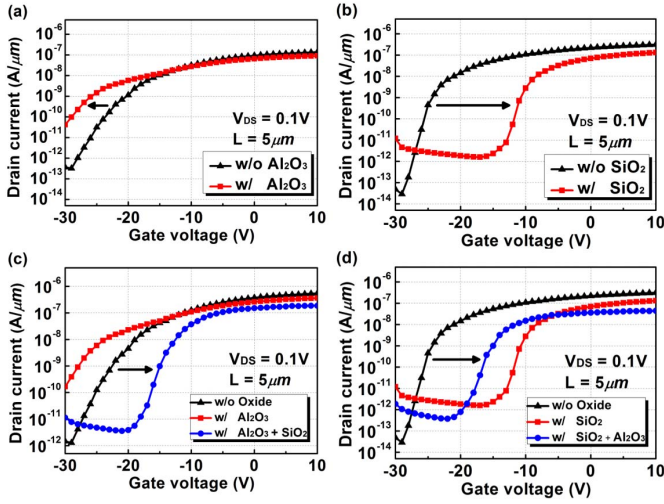


Fig. 2. (a), (b) The $I_d - V_g$ characteristics of a back-gated MoS₂ FET without/with single dielectric layer. (c), (d) The $I_d - V_g$ characteristics of a back-gated MoS₂ FET with a combination of two dielectric layers.

by using atomic force microscopy (AFM). After photoresist spin coating, photo-lithography was used to define the source/drain (S/D) regions. Ti (30nm) followed by Au (40nm) were deposited as metal contacts using electron beam evaporation at a pressure of 10^{-6} Pa. In the MoS₂ FET, the channel length is 5 μm and the width is 4 μm. The electrical properties of the MoS₂ FET were measured at room temperature using a Keithley 4200 Semiconductor Characterization System.

After the measurement, the dielectric layers of Al₂O₃ and SiO₂ were deposited on the MoS₂ FET. The dielectric layer was deposited using an electron beam evaporator at a pressure of 10^{-6} Pa. Since thin dielectric layer of Al₂O₃ deposited by atomic layer deposition (ALD) has low surface coverage on the MoS₂, electron beam evaporator was used [20]. The thickness of each layers was 5nm. Finally, the electrical measurement was performed again.

III. RESULTS AND DISCUSSIONS

To verify the effects of the dielectric layers, Al₂O₃ and SiO₂ dielectric capping layers were deposited individually, as shown in Fig. 1 (a) and (b), on separate MoS₂ FETs. Fig. 2 (a) and (b) show the $I_d - V_g$ characteristics of MoS₂ FETs with and without a single dielectric layer. The drain voltage was biased at 0.1V and the back gate voltage was applied from

−30V to 10V. The electrical characteristics depicted in the figures clearly show the change of device operation by the single dielectric layer. To numerically compare device operations, the V_{th} was extracted from the measured $I_d - V_g$ curve. According to the calculations, V_{th} of the MoS₂ FET changes by −3.56V with the Al₂O₃ dielectric layer and +9.20V with the SiO₂ dielectric layer. The V_{th} has no dependency on the thickness of each dielectric layer (not shown in here). This positive V_{th} shift with dielectric capping is a notable result for MoS₂-based FETs since previous research has only reported to induce negative V_{th} shifts in MoS₂-based FETs [21]. In addition to V_{th} , the field-effect carrier mobility is calculated by the formula; $\mu_e = L/(WV_dC_{OX}) \cdot (\partial I_d / \partial V_g)$ [11]. Following the single dielectric layer deposition, the mobility decreased from 27.3 cm²/V·s to 16.9 cm²/V.s with Al₂O₃ capping and from 49.9 cm²/V·s to 25.8 cm²/V.s with SiO₂ capping. The reduction of the mobility is caused by the defect charges in the dielectric layer deposited by e-beam evaporator, which has relatively more defects than the layer deposited by ALD [22], [23]. These defects induce carrier scattering in MoS₂ channel and decrease the carrier mobility of the device.

After determining the dielectric capping effects of single layers, we proceeded to deposit of double dielectric layers, one above the other, for the first time. Fig. 2 (c) and (d) show the $I_d - V_g$ characteristics of MoS₂ FETs with two different dielectric layers stacked in different orders. V_{th} shifts of +3.36V and +4.62V were induced by SiO₂ on Al₂O₃ and Al₂O₃ on SiO₂, respectively. In other words, positive V_{th} shifts were observed regardless of the stacking order. The carrier mobility is changed from 25.8cm²/V.s to 12.9cm²/V.s after dielectric capping with Al₂O₃ on SiO₂ and 16.9cm²/V.s to 15.7cm²/V.s after dielectric capping with SiO₂ on Al₂O₃.

In the previous study, the negative V_{th} shifts induced by high-k dielectric layer were explained by the presence of oxygen vacancies at the interface between the MoS₂ and the high-k dielectric layer [24]. While it is hard to explain positive V_{th} shifts induced by the single and double dielectric capping, the energy band diagram analysis based on the defect charges in the dielectric layer can explain the results shown in Fig. 2 [25]. Fig. 3 shows the band diagrams of MoS₂ FET with different dielectric layer stacks. The band diagrams on the left and the band diagrams on the right correspond to the y-direction and the x-direction in Fig. 1 (c), respectively. Since the typical operation of a MoS₂ device is explained by tuning the Schottky barrier between metal and MoS₂ [25], [26], both x- and y-directional investigations of band diagrams are required. In Fig. 3 (a), the positive defect charges existing in the Al₂O₃ interface cause the energy band of the MoS₂ interface to shift down [19]. Due to the band bending, the effective Schottky barrier between the contact metal and MoS₂ decreases, allowing the device to operate at a lower gate voltage than the one without dielectric capping. On the other hand, in Fig. 3 (b), the negative defect charges existing in the SiO₂ interface cause the energy band of the MoS₂ interface to shift up. Due to the band bending by the SiO₂ dielectric layer, the effective Schottky barrier between the contact metal and MoS₂ increases, indicating that the SiO₂ dielectric layer results in a positive shift of V_{th} . Moreover, if SiO₂ is stacked

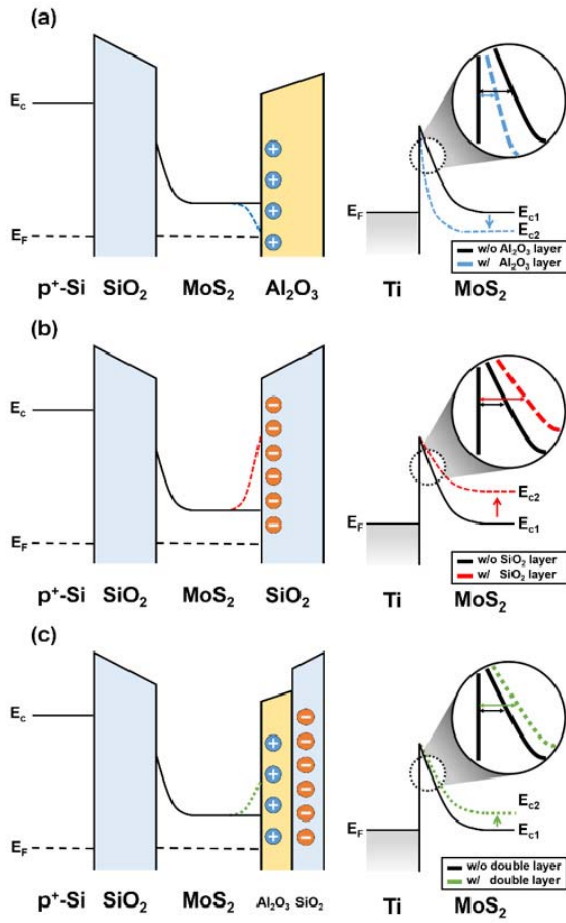


Fig. 3. The Energy band diagrams of MoS₂ FET affected from different directions by the type of dielectric layer (a) Al₂O₃, (b) SiO₂, (c) SiO₂ on Al₂O₃.

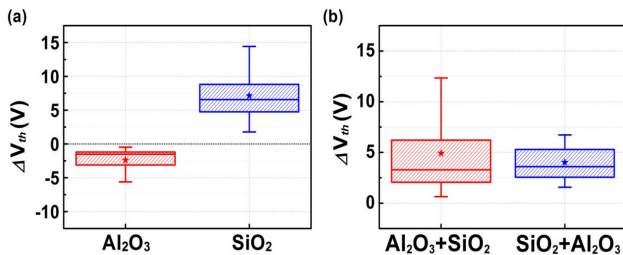


Fig. 4. (a) Average threshold voltage shifts by single dielectric layers. (b) Average threshold voltage shifts by differently ordered combinations of two dielectric layers.

on Al₂O₃, their defect charges partially cancel out each other, leaving weak negative net charges as shown in Fig. 3 (c). These remaining charges slightly shift up the energy band of the MoS₂ interface, resulting in a positive V_{th} shift. When Al₂O₃ is stacked on SiO₂, similar V_{th} shifts occurred. Combined with the electrical results shown in Fig. 2, these results imply that V_{th} shift in the MoS₂ FET is the sum of the defect charges in the stacked dielectric layers.

Fig. 4 (a) shows the box plots of V_{th} shift distribution of the MoS₂ FET devices with a single dielectric layer of Al₂O₃ and SiO₂, respectively. The results are extracted from ten devices for each dielectric layer. The average value of

V_{th} shifts was $-2.39V$ with Al₂O₃ capping and $+7.13V$ with SiO₂ capping. Fig. 4 (b) shows that MoS₂ FETs with either SiO₂ on Al₂O₃ or Al₂O₃ on SiO₂ show similar average V_{th} shifts, regardless of the deposited order. The average V_{th} shifts were $+4.92V$ with SiO₂ on Al₂O₃ and $+4.02V$ with Al₂O₃ on SiO₂. These results are consistent with the explanations of band bending described above.

IV. CONCLUSION

In summary, we have demonstrated the effect of single or double dielectric capping layers on a MoS₂ FET. Average V_{th} shifts of $-3.56V$ with Al₂O₃ capping and $+9.20V$ with SiO₂ capping were achieved. Additionally, we showed average V_{th} shifts of $+4.92V$ with the deposition of SiO₂ on Al₂O₃ and $+4.02V$ with the deposition of Al₂O₃ on SiO₂. This indicates that the defect charges in the dielectric layer induce band bending in the MoS₂ interface and adjust V_{th} of the MoS₂ FET. The proposed technique is a promising V_{th} adjustment technique for TMDC-based devices.

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